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NAAC GRADE A+

Worksheet 8

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Semester: 2nd Sem Date of Performance:12/01/2026

Subject Name: Technical Skills Subject Code: 25CAP-652_25MAM_KAR-1(A)

AIM: To apply the concept of Stored Procedures in database operations in order to perform tasks like insertion, updating, deletion, and retrieval of data efficiently, securely, and in a reusable manner within the database system.

S/W Requirement: Oracle Database Express Edition and pgAdmin

OBJECTIVES:

1. To understand the concept and usage of stored procedures in database systems.
2. To implement stored procedures for performing CRUD operations efficiently.
3. To enhance data security by restricting direct access through stored procedures.
4. To reduce code redundancy by using reusable database procedures.

To improve performance of database operations using precompiled stored procedures

Practical / Experiment Steps:

Step 0 Setup (Tables Structure-Employees)

```
CREATE TABLE staff (
   staff_id INT PRIMARY KEY,
   staff_name VARCHAR(50),
   salary NUMERIC(20,3)
);
INSERT INTO staff VALUES
(1, 'Rahul', 20000),
(2, 'Neha', 30000),
(3, 'Amit', 25000);
select * from staff;
```

	staff_id [PK] integer	staff_name character varying (50)	salary numeric (20,3)
1	1	Rahul	20000.000
2	3	Amit	25000.000
3	2	Neha	31000.000

```
-- Step 1: Create Procedure for Salary Increment
CREATE OR REPLACE PROCEDURE increment_salary_proc(
   IN P_STAFF_ID INT,
   INOUT P_INCREMENT NUMERIC(20,3),
   OUT RESULT_STATUS VARCHAR(25)
)
AS
$$
DECLARE
   NEW_SAL NUMERIC(20,3);
BEGIN
   -- Calculate new salary
   SELECT salary + P_INCREMENT
   INTO NEW_SAL
   FROM staff
   WHERE staff_id = P_STAFF_ID;
   -- If not found
   IF NOT FOUND THEN
      RAISE EXCEPTION 'STAFF NOT FOUND';
   END IF;
   -- Update salary
   UPDATE staff
   SET salary = NEW_SAL
   WHERE staff_id = P_STAFF_ID;
   -- Return values
   P_INCREMENT := NEW_SAL;
   RESULT_STATUS := 'UPDATED SUCCESSFULLY';
EXCEPTION
WHEN OTHERS THEN
```

```
    IF SQLERRM LIKE '%STAFF NOT FOUND%' THEN
        RESULT_STATUS := 'STAFF NOT FOUND';
    ELSE
        RESULT_STATUS := 'ERROR OCCURRED';
    END IF;
END;
$$ LANGUAGE plpgsql;
```

```
CREATE PROCEDURE

Query returned successfully in 95 msec.
```

```
DO
$$
DECLARE
    V_STAFF_ID INT := 2;
    V_STATUS VARCHAR(25);
    V_INCREMENT NUMERIC(20,3) := 1000;
BEGIN
 CALL increment_salary_proc(V_STAFF_ID, V_INCREMENT, V_STATUS);
    RAISE NOTICE
    'STATUS: % | UPDATED SALARY: %',
    V_STATUS, V_INCREMENT;
END;
$$;
```

```
NOTICE:  STATUS: UPDATED SUCCESSFULLY | UPDATED SALARY: 32000.000
DO

Query returned successfully in 68 msec.
```

Outcomes:

1. To demonstrate the ability to create and execute stored procedures for various database operations.
2. To apply stored procedures for performing secure and efficient CRUD operations.
3. To analyze the impact of stored procedures on database performance and execution efficiency.
4. To design modular and reusable database logic using stored procedures.